

Exam - Object-Oriented Programming

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Winter semester 2026/27 · 90 minutes · 105 points

Theory: 60 minutes (Tasks 1–5, 65 points). Programming: 30 minutes (Task 6, 40 points). **Instructor copy — do not distribute with the student paper.** Equivalent correct solutions earn full credit. Award partial credit by the criteria shown.

Name:	Student ID:
Date:	Signature:

Instructions

- Work individually. Answer Tasks 1–5 in the spaces provided; label any additional sheets with your student ID and task number.
- For Task 6, submit the Java source files in a folder named StudentID_LastName. Use a standard JDK (Java 17 or later), without preview features or external libraries.
- No GUI, interactive menu, file storage, or concurrency is required. Each task is independent.
- The invigilator will announce permitted aids and the submission channel before the exam starts.
- Check that your paper is complete using the page count in the footer.

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Task	1	2	3	4	5	6	Total
Available	10	15	16	12	12	40	105
Achieved							

THEORY · 60 MINUTES IN TOTAL

Task 1: Basic Concepts

10 points

For each situation, explain the relevant Java concept and give one short Java example. Each item earns **1 point for the explanation and 1 point for the example**.

a) Object references (2 points). Two variables should refer to the same Lamp object rather than two separate lamps. Explain what reference assignment copies.

Solution / marking

Assignment copies the reference, not the object (1). Example: `Lamp a = new Lamp(); Lamp b = a;` (1). Changes to that object's state are observable through either reference.

b) Encapsulation (2 points). A thermostat must reject target temperatures outside its allowed range. Explain why its temperature field should not be public.

Solution / marking

A private field prevents callers from bypassing validation (1). Example: `private int target;` with `public void setTarget(int t) { if (t < 5 || t > 30) throw new IllegalArgumentException(); target = t; }` (1).

c) Constructors (2 points). A Room must receive its name when it is created. Explain the constructor's role and show the constructor declaration.

Solution / marking

The constructor initializes a new instance (1). Example inside Room: `private String name; public Room(String name) { this.name = name; }` (1). A constructor has the class name and no return type.

d) Static members (2 points). All Room objects share the same maximum permitted occupancy. Explain why a class constant is suitable.

Solution / marking

A static member belongs to the class rather than each instance (1). Example: `public static final int MAX_OCCUPANCY = 20;` (1). `final` prevents reassignment.

e) Immutable strings (2 points). Explain why calling `name.toUpperCase()` alone does not change the string stored in `name`. Show how to retain the result.

Solution / marking

Strings are immutable; the method returns a string rather than modifying the original (1). `name = name.toUpperCase();` (1).

THEORY · 60 MINUTES IN TOTAL

Task 2: Object-Oriented Reasoning

15 points

a) Inheritance and super (5 points). Given the superclass below, write a subclass `Bicycle` with a constructor `Bicycle(String id)` that initializes the inherited state. Override `label()` to return `Bike:` followed by the superclass `label`. Keep the field `private` and use `super` in both places.

```
1  class Vehicle {
2      private String id;
3      public Vehicle(String id) { this.id = id; }
4      public String label() { return id; }
5  }
```

Java

Solution / marking

```
1  class Bicycle extends Vehicle {
2      public Bicycle(String id) { super(id); }
3      @Override
4      public String label() { return "Bike: " + super.label(); }
5  }
```

Java

Subclass declaration (1); constructor and `super(id)` (2); public override with the required result and `super.label()` (2).

b) Overloading and overriding (5 points). Read this separate program. State its two output lines (2 points). Identify the overloaded methods (1 point), the overridden method (1 point), and explain why the first line uses `Bell.sound()` (1 point).

```
1  class Signal {
2      public String sound() { return "beep"; }
3      public String sound(int count) { return "count=" + count; }
4  }
5  class Bell extends Signal {
6      @Override
7      public String sound() { return "ding"; }
8  }
9  // Inside main:
10 Signal signal = new Bell();
11 System.out.println(signal.sound());
12 System.out.println(signal.sound(2));
```

Java

Solution / marking

Output: `ding`, then `count=2` (1 each). `Signal.sound()` and `Signal.sound(int)` are overloaded (1). `Bell.sound()` overrides `Signal.sound()` (1). Runtime dispatch uses the actual object's type, `Bell`, for the no-argument instance method (1).

c) Interface or abstract class? (5 points). Unrelated classes must all offer `double energyUse()`. Give a suitable interface declaration (2 points). A family of appliances additionally shares a private `name` field, a constructor, and a concrete getter. Explain why an abstract superclass can hold this shared implementation (2 points). Can that abstract superclass be instantiated directly (1 point)?

Solution / marking

`interface EnergyUser { double energyUse(); }` (2). An abstract class can contain instance state (1), constructors and implemented methods shared by subclasses (1). It cannot be instantiated directly (1).

THEORY · 60 MINUTES IN TOTAL

Task 3: True or False

16 points

Mark exactly one column for each statement. Each correct answer earns **2 points**. An incorrect, missing, or double-marked answer earns 0 points; there are no deductions.

No.	Statement	True	False
1	A local variable of a class type stores a reference to an object, or <code>null</code> .	X	
2	Declaring an array reference <code>final</code> prevents changes to its elements.		X
3	A subclass inherits its superclass's constructors.		X
4	A class can implement two interfaces while extending one class.	X	
5	A static method can read an instance field if it has a suitable object reference and access is permitted.	X	
6	<code>String.equals</code> compares the text content of two strings.	X	
7	The first valid index of a nonempty Java array is 1.		X
8	A method declared <code>void</code> must end with <code>return null;</code> .		X

Solution / marking

1. True: a class-typed variable stores a reference value.
2. False: `final` prevents reassignment of the reference, not mutation of the array.
3. False: constructors are not inherited.
4. True: Java permits multiple implemented interfaces and one direct superclass.
5. True: static methods can use an accessible field via an object.
6. True: `equals` compares string contents.
7. False: indices start at 0.
8. False: a void method can fall through or use `return;`, not return a value.

THEORY · 60 MINUTES IN TOTAL

Task 4: Code Completion

12 points

Complete gaps **A-F** directly in the listing (**2 points per gap**). WaterTank stores an integer water level from **0 through 100 inclusive**. Invalid constructor or setter arguments must throw `IllegalArgumentException` without changing an existing level. The constructor must reuse the setter. `freeCapacity()` returns `100 - level`. The string format is exactly `WaterTank[level=25]` when the level is 25.

```
1      public class WaterTank {
2          // A: declare the private int field
3          -----
4
5      public WaterTank(int level) {
6          // B: initialize through the setter
7          -----
8      }
9      public int getLevel() {
10         // C
11         -----
12     }
13     public void setLevel(int level) {
14         // D: validate first, then update the field
15         -----
16         -----
17         -----
18         -----
19     }
20     public int freeCapacity() {
21         // E
22         -----
23     }
24     @Override
25     public String toString() {
26         // F
27         -----
28     }
29 }
```

Java

Solution / marking

```
1    public class WaterTank {
2        private int level;
3        public WaterTank(int level) { setLevel(level); }
4        public int getLevel() { return level; }
5        public void setLevel(int level) {
6            if (level < 0 || level > 100) {
7                throw new IllegalArgumentException();
8            }
9            this.level = level;
10       }
11       public int freeCapacity() { return 100 - level; }
12       @Override
13       public String toString() {
14           return "WaterTank[level=" + level + "];"
15       }
16   }
```

Java

A: correct type/name (1), private (1). B: calls setLevel (1) with its parameter (1). C: returns the field (2). D: rejects both invalid ranges before assignment (1), assigns valid input (1). E: correct result (2). F: exact string using the field (2).

THEORY · 60 MINUTES IN TOTAL

Task 5: Code Debugging

12 points

Find **six independent defects**. For each, identify the location, explain the problem, and give a correction (**1 point for the explanation, 1 for the correction**). The specification requires all instance fields to be private; the constructor to retain its arguments; `hasName` to compare text content; and `sum` to add every array element exactly once. Assume names and arrays are non-null. An empty array must sum to 0. Missing getters, formatting choices, and Java's default field initialization are **not** defects.

```
1      public class ScoreCard {
2          public String name;
3          private int bonus;
4
5          public ScoreCard(String name, int bonus) {
6              this.name = name
7              bonus = bonus;
8          }
9          public static int getBonus() {
10             return bonus;
11         }
12         public boolean hasName(String other) {
13             return name == other;
14         }
15         public int sum(int[] scores) {
16             int total = 0;
17             for (int i = 0; i ≤ scores.length; i++) {
18                 total += scores[i];
19             }
20             return total;
21         }
22     }
```

Java

Solution / marking

1. name is public, contrary to the specification; declare it private.
2. this.name = name lacks a semicolon; add ;.
3. bonus = bonus assigns the parameter to itself; use this.bonus = bonus;.
4. getBonus is static but accesses an instance field without an object; remove static.
5. = tests reference equality; use name.equals(other).
6. i ≤ scores.length accesses index scores.length; use i < scores.length.

Corrected reference implementation:

```
1  public class ScoreCard {
2      private String name;
3      private int bonus;
4      public ScoreCard(String name, int bonus) {
5          this.name = name;
6          this.bonus = bonus;
7      }
8      public int getBonus() { return bonus; }
9      public boolean hasName(String other) {
10         return name.equals(other);
11     }
12     public int sum(int[] scores) {
13         int total = 0;
14         for (int i = 0; i < scores.length; i++) {
15             total += scores[i];
16         }
17         return total;
18     }
19 }
```

Java

PROGRAMMING · 30 MINUTES

Task 6: Room Sensor Log

40 points

Build a small console application that stores room measurements and calculates a summary. Use three files: `Reading.java`, `SensorLog.java`, and `Demo.java`. You do **not** need to implement keyboard input, editing, searching, or deletion.

Data and behavior

- A `Reading` has a room name (`String`) and an integer temperature in °C (`int`). Once created, it has no methods that change these values.
- The `Reading` constructor receives valid values; validation is the responsibility of `SensorLog.addReading`.
- A valid room name is non-null and not empty after `trim()`. Store the trimmed name. Temperatures must be between **-20 and 50 inclusive**.
- Repeated room names are allowed: they represent separate measurements.
- `addReading(String room, int temperature)` returns `true` after adding a valid reading. Otherwise it returns `false` and leaves the list unchanged.
- `countAbove(int limit)` returns the number of readings **strictly greater than** the limit.
- `averageTemperature()` returns a double average over all stored readings. It returns `0.0` when there are none. Avoid integer division.
- `displayAll()` prints each reading in insertion order, or `No readings` when empty.

Implementation and points

Part	Deliverable	Points
a	Create the named project folder and the three Java files with matching public classes.	3
b	<code>Reading</code> : two private fields, constructor, both getters, and a readable <code>toString()</code> containing both values.	5
c	<code>SensorLog</code> : private <code>ArrayList<Reading></code> initialized as an empty list.	4
d	Implement boolean <code>addReading(String room, int temperature)</code> with the specified validation.	6
e	Implement <code>int countAbove(int limit)</code> ; an empty log returns 0.	5
f	Implement <code>double averageTemperature()</code> including its empty-log behavior.	6
g	Implement <code>void displayAll()</code> including the empty message.	3
h	<code>Demo.main</code> : demonstrate the acceptance checks on the next page with labelled output or assertions.	5
i	Use clear names, consistent indentation, and encapsulation; keep log operations in <code>SensorLog</code> , not <code>Demo</code> .	3

Task 6: Acceptance checks and API reminder

Use a fresh log for this sequence:

1. Display it while empty; check that its average is 0.0 and countAbove(20) is 0.
2. Add (" Kitchen ", 20) and ("Kitchen", 21); both calls must return true. The displayed room names must have no surrounding spaces.
3. Check that the average is 20.5, countAbove(20) is 1, and countAbove(21) is 0.
4. Try a null room name, a spaces-only room name, and a temperature of 51; all must return false. The average and counts must remain unchanged.
5. Display the two valid readings in their insertion order.

ArrayList needs `import java.util.ArrayList;`. These operations may help:

```
1  ArrayList<Reading> readings = new ArrayList<>();
2  readings.add(new Reading("Hall", 18));
3  int size = readings.size();
4  boolean empty = readings.isEmpty();
5  for (Reading reading : readings) {
6      System.out.println(reading);
7  }
8  // For non-null text:
9  boolean blank = text.trim().isEmpty();
```

Java

Submit all three .java files. A compiling partial solution can earn credit for the parts it implements correctly.

Solution / marking

Reading.java

```
1 public class Reading {
2     private final String room;
3     private final int temperature;
4     public Reading(String room, int temperature) {
5         this.room = room;
6         this.temperature = temperature;
7     }
8     public String getRoom() { return room; }
9     public int getTemperature() { return temperature; }
10    @Override
11    public String toString() { return room + ": " + temperature + " C"; }
12 }
```

Java

SensorLog.java

```
1 import java.util.ArrayList;
2 public class SensorLog {
3     private final ArrayList<Reading> readings = new ArrayList<>();
4     public boolean addReading(String room, int temperature) {
5         if (room == null || room.trim().isEmpty()
6             || temperature < -20 || temperature > 50) {
7             return false;
8         }
9         readings.add(new Reading(room.trim(), temperature));
10        return true;
11    }
12    public int countAbove(int limit) {
13        int count = 0;
14        for (Reading r : readings) {
15            if (r.getTemperature() > limit) count++;
16        }
17        return count;
18    }
19    public double averageTemperature() {
20        if (readings.isEmpty()) return 0.0;
21        double sum = 0;
22        for (Reading r : readings) sum += r.getTemperature();
23        return sum / readings.size();
24    }
25    public void displayAll() {
26        if (readings.isEmpty()) System.out.println("No readings");
27        for (Reading r : readings) System.out.println(r);
28    }
29 }
```

Java

Demo.java (run with `java -ea Demo` to enable assertions).

```
1  public class Demo {
2      public static void main(String[] args) {
3          SensorLog log = new SensorLog();
4          log.displayAll();
5          assert log.averageTemperature() == 0.0;
6          assert log.countAbove(20) == 0;
7          assert log.addReading(" Kitchen ", 20);
8          assert log.addReading("Kitchen", 21);
9          assert log.averageTemperature() == 20.5;
10         assert log.countAbove(20) == 1;
11         assert log.countAbove(21) == 0;
12         assert !log.addReading(null, 20);
13         assert !log.addReading("  ", 20);
14         assert !log.addReading("Hall", 51);
15         assert log.averageTemperature() == 20.5;
16         assert log.countAbove(20) == 1;
17         assert log.countAbove(21) == 0;
18         log.displayAll();
19     }
20 }
```

Java

Marking (40 points): a: folder (1), classes/files (2). b: private fields (1), constructor (1), getters (2), string (1). c: private typed list (2), empty initialization (2). d: name validation including null (2), bounds (1), trim (1), add/true (1), unchanged/false (1). e: traversal (2), strict comparison (2), count/empty return (1). f: empty guard (2), sum (2), floating-point division (2). g: all readings in order (2), empty message (1). h: one point for each of the five acceptance checks. i: naming (1), formatting (1), responsibility separation/encapsulation (1). Do not deduct twice for one underlying error.